Software Requirements Specification

for

SmartClass

Integrated Academic and Classroom Management Platform

Version 1.0 (Approved)

Prepared by: Baibhav Pati (47)
Lavanya Gurbani (16)
Anvi Patil (48)

Document Date: 13 August 2026

Document structure adapted from the IEEE-style Software Requirements Specification template (Wiegers, Software Requirements, 2nd ed.)

Revision History

Name	Date	Reason for Changes	Version
SmartClass Requirements Team	13-Aug-2026	Initial baseline SRS	1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for Version 1.0 of SmartClass, an integrated academic and classroom management platform for a college environment. It specifies the complete requirements baseline covering authentication, class/course administration, attendance, lecture and timetable management, classroom availability, study material, assignments, quizzes, results, dashboards, communication, calendar, council events, discussion forums, grievance handling, and AI-generated test performance reporting. This document covers the entire SmartClass system as a single, self-contained web application; it does not describe a subsystem of a larger product.

1.2 Document Conventions

Requirement statements use “shall” to denote a mandatory, verifiable capability of the system. Each functional requirement carries a unique identifier of the form FR-<AREA>-<NNN> (e.g., FR-ATT-003); nonfunctional requirements use NFR-<AREA>-<NNN>; business rules use BR-<NNN>; unresolved items use TBD-<NNN>. Every requirement in Section 4 inherits the priority (High, Medium, or Low) stated for its parent feature unless individually noted otherwise. Bulleted lists under “Stimulus/Response Sequences” describe representative user–system interaction flows rather than exhaustive requirement statements; the authoritative, verifiable statements are the numbered requirements that follow each such list.

1.3 Intended Audience and Reading Suggestions

This document is intended for the SmartClass project sponsor, the development team (frontend, backend, and database engineers), quality assurance/test engineers, UI/UX designers, and academic stakeholders (college administration) who will validate that the specified behavior matches institutional needs.

Readers new to the project should begin with Section 2 (Overall Description) for context, then proceed to Section 4 (System Features) for the detailed functional baseline relevant to their area of work. Developers implementing a specific feature should read the corresponding subsection of Section 4 together with the relevant Business Rules in Section 5.5 and the Glossary in Appendix A. Quality assurance engineers should additionally review Section 5 (Nonfunctional Requirements) and Appendix B (Analysis Models) when constructing verification and validation plans. This SRS deliberately does not define software architecture, class structures, design patterns, or database schema — those are addressed in a separate Software Architecture and Design document that will be produced downstream of this baseline.

1.4 Product Scope

SmartClass is a responsive, full-stack web application (not a native Android or iOS application) that centralizes college academic and classroom-management activity for four user categories: Student, Faculty, Admin, and Council Head, with an additional HOD permission set that may be combined with the Faculty role. The platform's objective is to replace fragmented manual and ad hoc digital processes — paper attendance registers, scattered notices, disconnected spreadsheets for timetabling — with a single system through which academic and institutional information relevant to each role is created, tracked, and consumed.

In scope for Version 1.0: authentication and role-based access control; class and course administration; manual, lecture-level attendance with Regular/Substitute/Extra lecture support; complete and day-wise timetable views with conflict detection; free-classroom search; study material distribution; assignment creation, submission, and grading; online objective quizzes with automatic evaluation; results and performance viewing; role-specific dashboards; announcements and notifications; a filterable calendar; college council event management with automatic calendar/announcement generation; per-course discussion forums; a feedback/grievance workflow; and AI-generated, advisory test-performance reports. Explicitly out of scope for Version 1.0 are the items enumerated in Section 2.5 and Section 6.

1.5 References

- Wiegers, Karl E. Software Requirements Specification template, © 1999 (structural and organizational reference for this document).
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications (structural reference for content organization).

No additional external standards, contracts, or vision-and-scope documents were supplied at the time of writing. Where a reference would ordinarily be required but its bibliographic details cannot be verified from the source material provided for this SRS, the item has been placed in Appendix C (To Be Determined List) rather than fabricated.

2. Overall Description

2.1 Product Perspective

SmartClass is a new, self-contained product; it is not a follow-on to an existing system and does not replace a specific legacy platform identified at the time of writing. It is conceptually comparable to general-purpose classroom platforms such as Google Classroom in its handling of courses, material, and assignments, but is purpose-built as an integrated college academic platform with functionality those general-purpose tools do not provide out of the box: manual lecture-level attendance with Regular/Substitute/Extra lecture tracking, timetable and classroom-availability management, college council event publishing, a grievance workflow, and AI-generated advisory test-performance reports.

The system comprises a browser-based client, a backend application/API layer, a relational database, file storage for uploaded material and submissions, and an external AI/LLM service used exclusively for the advisory test-performance report described in Section 4.17. Figure 1 references to a system context diagram and component diagram are identified in Appendix B as analysis artifacts to be produced separately.

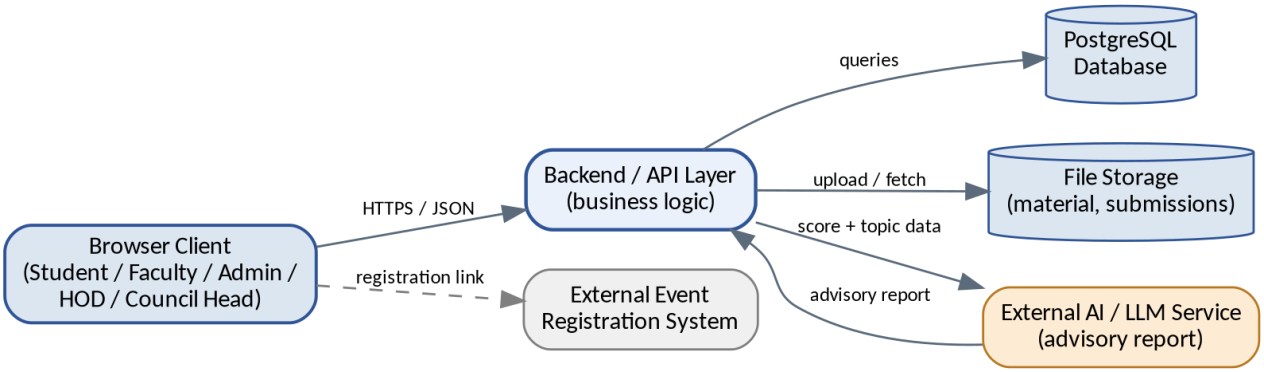


Figure 1 — System context: client, backend, data stores, and external services

2.2 Product Functions

At a high level, SmartClass lets users:

- Authenticate securely and operate within a role-based permission model (Student, Faculty, Admin, Council Head, and an HOD permission combinable with Faculty).
- Administer the academic structure: classes, courses, and their faculty associations (Admin).
- Record and review manual, lecture-level attendance, including Regular, Substitute, and Extra lecture types, with automatic percentage calculation and low-attendance alerting.
- View both complete and day-wise timetables, and search for free classrooms for a given date/time.
- Distribute and access study material, create and submit assignments, and create and attempt objective online quizzes with automatic evaluation.
- View results and role-appropriate academic performance analytics through dedicated dashboards.
- Publish and receive announcements and notifications, participate in per-course discussion forums, and use a filterable academic calendar.

- Create and publish college council events with automatic calendar and announcement generation.
- Submit, track, and resolve feedback/grievances through a defined status lifecycle.
- Receive AI-generated, advisory reports identifying strengths and weaknesses after a quiz/test is officially scored.

2.3 User Classes and Characteristics

User Class	Characteristics
Student	Highest-frequency user class. Receives credentials from Admin; assigned to exactly one class at a time. Primarily consumes academic content (material, assignments, quizzes, results, attendance, timetable) and participates in forums and grievance submission. Assumed to have basic web-browser literacy, frequently on mobile devices.
Faculty	Regular, class/course-scoped user. Receives credentials from Admin. Creates and manages academic content for assigned classes/courses, records attendance, and may teach the same course across multiple classes. May additionally hold the HOD permission.
HOD (Faculty + HOD permission)	Not a separate account type. A Faculty user granted additional departmental permissions: departmental analytics and grievance handling, layered on top of standard Faculty capability.
Admin	Highest system authority. Configures the academic structure, creates all other accounts, manages classrooms and timetable data, and administers grievances and institutional information. Lower frequency of use than Student/Faculty but broadest scope.
Council Head	Narrow-scope user responsible only for creating and managing events for one college council. Receives credentials from Admin. Infrequent, event-driven usage pattern.

The diagram below summarizes how Admin provisions each downstream role and how the HOD permission layers onto an existing Faculty account rather than creating a separate identity.

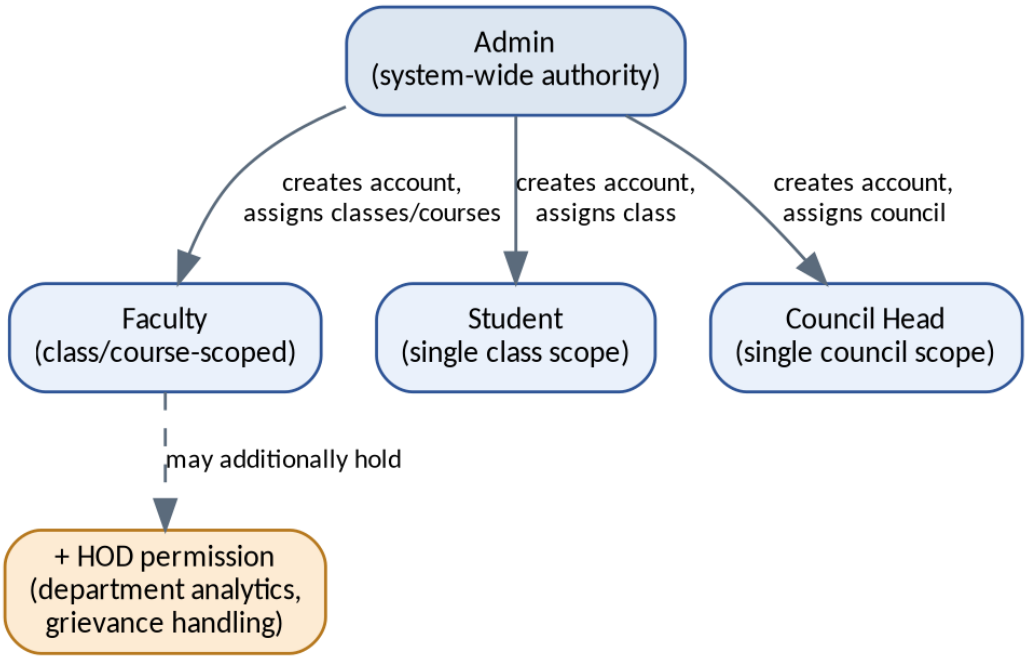


Figure 2 — Role provisioning and the Faculty/HOD permission relationship

2.4 Operating Environment

SmartClass is delivered as a responsive web application accessed through modern desktop and mobile web browsers (current major versions of Chrome, Firefox, Safari, and Edge) on desktop/laptop computers, tablets, and smartphones. No native Android or iOS application is provided. Server-side components run on a backend/API tier backed by a PostgreSQL relational database and file storage for uploaded documents and submissions, as described further in Section 3.3 and Section 16 (Technology Context) of the accompanying project documentation. No specialized client hardware (e.g., QR scanners or biometric readers) is required.

2.5 Design and Implementation Constraints

- The system shall be implemented as a responsive web application; a native mobile application is out of scope for Version 1.0.
- Attendance shall be recorded manually by Faculty; QR-code, biometric, and facial-recognition attendance mechanisms are explicitly out of scope.
- The system shall use a relational database (PostgreSQL) for persistent storage of academic, attendance, and administrative data.
- The system shall communicate with client browsers exclusively over HTTPS.
- Passwords shall never be stored or transmitted in plaintext (see NFR-SEC requirements in Section 5.3).
- The AI-generated test performance report is advisory only; official academic scores shall always be produced by deterministic evaluation logic, independent of AI service availability.
- The SRS specifies business requirements, not database schema or software architecture; specific design patterns, ORM choices, and API endpoint design are addressed in downstream architecture and design documentation.

2.6 User Documentation

The following user-facing documentation is planned to accompany the delivered software: a Student quick-start guide, a Faculty quick-start guide covering attendance and academic content management, an Admin configuration guide covering class/course/user setup, and role-appropriate in-application help/tooltip content. Delivery format (online help pages versus downloadable PDF guides) is recorded as TBD-002 in Appendix C.

2.7 Assumptions and Dependencies

- It is assumed that Admin will provision accurate class, course, and faculty-assignment data before Students and Faculty begin using the system; incorrect setup data will propagate incorrect access scoping.
- It is assumed that each institution using SmartClass will configure a low-attendance threshold appropriate to its own academic policy.
- The system depends on an external AI/LLM service for the advisory test-performance report; the specific vendor is not mandated by this SRS and is treated as a configurable, replaceable dependency (see Section 3.3).
- The system depends on external event-registration platforms referenced by council-event registration links; SmartClass does not implement event registration itself.
- It is assumed that end users have consistent access to a modern web browser and an internet connection of sufficient quality for typical academic web application usage.

3. External Interface Requirements

3.1 User Interfaces

SmartClass shall provide a responsive, browser-based graphical user interface covering, at minimum, the following logical screens/areas: login and first-login password change; Student, Faculty, Admin, HOD, and Council dashboards; class and course views; study material browsing and upload; attendance marking and attendance history; complete and day-wise timetable views; free-classroom search; assignment creation, submission, and grading; quiz creation and attempt; results and performance views; AI-generated test report views; announcements and notifications; calendar (Events/Tests/Holidays); discussion forums; grievance submission and tracking; and council event creation and browsing. Every screen shall present role-consistent navigation and a standard mechanism for accessing help and logging out. Detailed visual design, wireframes, and interaction specifications are outside the scope of this SRS and will be documented separately in a UI design specification; this document defines only the logical presence and behavior of each interface area.

3.2 Hardware Interfaces

SmartClass requires no specialized hardware. It operates through ordinary desktop computers, laptops, tablets, and smartphones equipped with a supported web browser. No QR-code scanners, biometric capture devices, or other specialized attendance hardware are required or supported, consistent with the manual attendance model defined in Section 4.3.

3.3 Software Interfaces

Interface	Description
Web Browser	Primary client interface; the system exchanges HTML/JS/CSS and JSON data with the browser over HTTPS.
Backend / API Layer	Application layer exposing the business logic and data operations consumed by the browser client.
PostgreSQL Database	System of record for classes, courses, users, attendance, lectures, timetable, assignments, quizzes, results, announcements, calendar, events, forums, and grievances.
File Storage	Stores uploaded study material, assignment attachments, and submission files; referenced by the backend/API layer.
External AI/LLM Service	Receives deterministic score and topic-wise performance data for a completed quiz/test and returns an advisory performance analysis, per Section 4.17. Specific vendor is a configurable implementation detail, not a fixed requirement of this SRS.
External Event Registration Links	Council event records may reference an externally hosted registration system via a URL; SmartClass does not implement registration processing.

3.4 Communications Interfaces

- All client–server communication shall occur over HTTPS.

- The backend/API layer shall expose REST-style (or equivalent) API operations consumed by the browser client.
- Communication with the external AI/LLM service (Section 4.17) shall occur over an encrypted, authenticated channel.
- Database and file-storage communication shall occur over a secure, authenticated channel not directly exposed to end users.
- Email or other outbound notification-channel integration is not mandated by this SRS; in-application notifications (Section 4.12) satisfy the baseline requirement. Email delivery, if adopted, will be documented as an implementation-level addition.

4. System Features

This section specifies SmartClass's functional requirements organized by system feature, the major services the product provides. Each feature states its description and priority, representative stimulus/response sequences, and the detailed, uniquely identified functional requirements that a development team can implement and a test team can verify directly.

4.1 Authentication and Access Control

4.1.1 Description and Priority

Provides secure identity verification and role-based authorization for all SmartClass user categories (Student, Faculty, Admin, Council Head), including support for a combined Faculty/HOD permission set. This feature underlies every other feature in the system, since all functionality is gated by an authenticated session and the caller's assigned role(s).

Priority: High.

4.1.2 Stimulus/Response Sequences

- Admin creates a Student, Faculty, or Council Head account and the system generates a unique ID and temporary password, which Admin communicates to the user.
- A user submits an ID and password on the login screen; the system validates the credentials and, on success, establishes a session and routes the user to the role-appropriate dashboard.
- A user logging in for the first time with a temporary password is prompted to set a new password before proceeding.
- A user attempts to access a function or resource outside their authorized role or scope; the system denies the request and logs the attempt.
- A user selects "log out"; the system terminates the session/token.

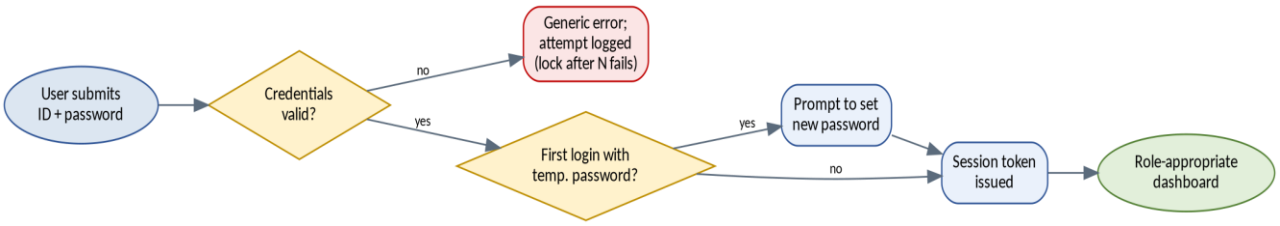


Figure 3 — Login and session-establishment flow

4.1.3 Functional Requirements

ID	Requirement
FR-AUTH-001	The system shall allow a registered user (Student, Faculty, Admin, or Council Head) to authenticate using a unique user ID and password.
FR-AUTH-002	The system shall allow Admin to create Student, Faculty, and Council Head accounts, each with a system-generated unique ID and a temporary password.

ID	Requirement
FR-AUTH-003	The system shall require a user to change a temporary password to a new password meeting the configured password policy before accessing any other function, upon first successful login.
FR-AUTH-004	The system shall store all passwords using a one-way cryptographic hash with per-password salt; the system shall never store or transmit passwords in plaintext.
FR-AUTH-005	The system shall reject authentication attempts with invalid credentials and shall display a generic error message that does not reveal whether the ID or the password was incorrect.
FR-AUTH-006	The system shall lock an account for a configurable cool-down period after a configurable number of consecutive failed login attempts, and shall log the lockout event.
FR-AUTH-007	The system shall issue a time-limited session token upon successful authentication and shall invalidate it upon logout, expiry, or password change.
FR-AUTH-008	The system shall enforce role-based access control such that a user can only invoke functions and view data permitted to their assigned role(s).
FR-AUTH-009	The system shall support a user holding more than one role concurrently (e.g., Faculty and HOD) and shall present the combined set of permissions for that user without requiring a separate account.
FR-AUTH-010	The system shall deny access and log the event when a user attempts to access a function or record outside their authorized scope.
FR-AUTH-011	The system shall allow an authenticated user to log out, terminating the active session/token immediately.
FR-AUTH-012	The system shall allow a user to change their own password after authenticating with their current password.

4.2 Class and Course Management

4.2.1 Description and Priority

Enables Admin to define the academic structure that all other features depend on: classes (student batches/divisions), courses (subjects), and the associations between classes, courses, and the faculty assigned to teach them. On login, Faculty see the classes and courses assigned to them, and Students see their assigned class and its associated courses.

Priority: High.

4.2.2 Stimulus/Response Sequences

- Admin creates a new class (e.g., "TE Computer Engineering – Division A") and subsequently adds students to it.
- Admin creates or edits a course record and associates it with one or more classes.

- Admin assigns a faculty member to teach a specific course for a specific class; the system supports the same faculty member being assigned the same course across multiple classes.
- A Faculty user logs in and the system displays the classes assigned to them and, within each class, the courses they are assigned to teach.
- A Student user logs in and the system displays their assigned class and the courses associated with that class.

4.2.3 Functional Requirements

ID	Requirement
FR-CLASS-001	The system shall allow Admin to create, edit, and deactivate a class record identifying a student batch/division.
FR-CLASS-002	The system shall allow Admin to add and remove students from a class.
FR-CLASS-003	The system shall allow Admin to create, edit, and deactivate a course record.
FR-CLASS-004	The system shall allow Admin to associate one or more courses with a class.
FR-CLASS-005	The system shall allow Admin to assign a faculty member to teach a specific course for a specific class.
FR-CLASS-006	The system shall support assigning the same faculty member to teach the same course to more than one class.
FR-CLASS-007	Upon Faculty login, the system shall display only the classes and courses to which that faculty member is currently assigned.
FR-CLASS-008	Upon Student login, the system shall display only the student's assigned class and the courses associated with that class.
FR-CLASS-009	The system shall prevent a Student or Faculty user from viewing or modifying class/course data for which they are not authorized.
FR-CLASS-010	The system shall allow Admin to reassign a student to a different class, preserving the student's historical academic records under the prior class association.

4.3 Attendance Management

4.3.1 Description and Priority

Provides manual, lecture-level attendance recording by Faculty and attendance viewing, statistics, and low-attendance alerting for Students, Faculty, HOD, and Admin. Attendance is recorded exclusively through manual faculty action; the system does not use QR codes, biometric devices, or facial recognition.

Priority: High.

4.3.2 Stimulus/Response Sequences

- Faculty selects a class and a scheduled lecture/session, and the system displays the enrolled student list for that lecture.

- Faculty marks each listed student Present or Absent and saves the attendance; the system stores one attendance record per student per lecture.
- A Student views their attendance history and the system displays a lecture-by-lecture record together with subject-wise and overall percentages.
- The system recomputes a student's attendance percentage after each new attendance record is saved and compares it against the configured institutional threshold.
- A student's attendance percentage falls below the configured threshold; the system generates a low-attendance alert visible to the student and, in aggregate, to the relevant Faculty and HOD as a defaulter listing.

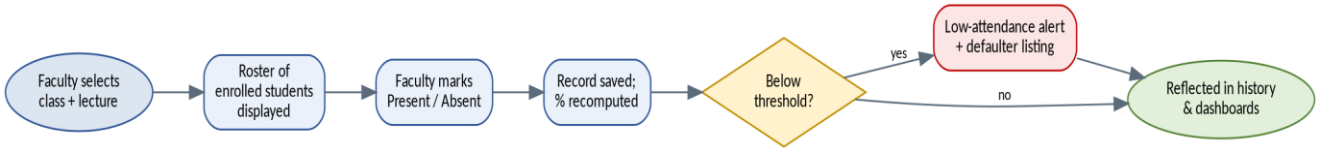


Figure 4 — Attendance recording and low-attendance alerting flow

4.3.3 Functional Requirements

ID	Requirement
FR-ATT-001	The system shall allow an authorized Faculty user to select a class and a specific lecture/session for which to record attendance.
FR-ATT-002	The system shall display the roster of students enrolled in the selected class for the selected lecture.
FR-ATT-003	The system shall allow Faculty to mark each student as Present or Absent for the selected lecture.
FR-ATT-004	The system shall persist attendance as one record per student per lecture/session upon Faculty save action.
FR-ATT-005	The system shall not provide any QR-code-based, biometric, or facial-recognition mechanism for recording attendance.
FR-ATT-006	The system shall allow Faculty to edit previously saved attendance for a lecture within a configurable correction window, and shall retain an audit trail of such changes.
FR-ATT-007	The system shall allow a Student to view their own attendance history, including subject-wise and overall attendance percentage.
FR-ATT-008	The system shall allow Faculty to view attendance history and statistics for classes/courses they are authorized to teach.
FR-ATT-009	The system shall calculate attendance percentage using only valid conducted lectures (Regular, Substitute, and Extra), excluding cancelled lectures.
FR-ATT-010	The system shall identify attendance defaulters as students whose attendance percentage falls below the configured institutional threshold.

ID	Requirement
FR-ATT-011	The system shall generate a low-attendance alert to a Student when their attendance percentage crosses below the configured threshold.
FR-ATT-012	The system shall allow Faculty to view a defaulter list for their authorized classes/courses.
FR-ATT-013	The system shall prevent Students from creating, modifying, or deleting any attendance record.
FR-ATT-014	The system shall prevent a Faculty member from recording or viewing attendance for a class/course to which they are not assigned.

4.4 Lecture and Session Management

4.4.1 Description and Priority

Manages the lecture/session records to which attendance is attached, including Regular, Substitute, and Extra lecture types. This feature preserves the distinction between the originally scheduled faculty member and the faculty member who actually conducted a substituted lecture, and ensures extra lectures are fully integrated into scheduling and attendance calculations.

Priority: High.

4.4.2 Stimulus/Response Sequences

- The scheduled Regular lecture occurs as timetabled and Faculty records attendance against it directly.
- The originally assigned faculty member for a scheduled lecture is unavailable; an authorized user records a Substitute lecture, preserving both the originally scheduled faculty and the actual conducting faculty, and attendance is recorded against the actual conducted lecture.
- An authorized user creates an Extra lecture outside the normal timetable, specifying class, course, faculty, date, time, and classroom; the lecture becomes available for attendance and is reflected in scheduling views.
- A user views attendance history and the system displays, for each entry, the lecture type (Regular, Substitute, or Extra).
- An authorized user cancels a scheduled lecture; the system excludes the cancelled lecture from attendance calculations.

The state model below shows how a lecture record moves between Scheduled, Substitute, Extra, Conducted, and Cancelled states, and clarifies that attendance is always recorded against the Conducted state regardless of which path produced it — this is the state-model artifact identified in Appendix B.

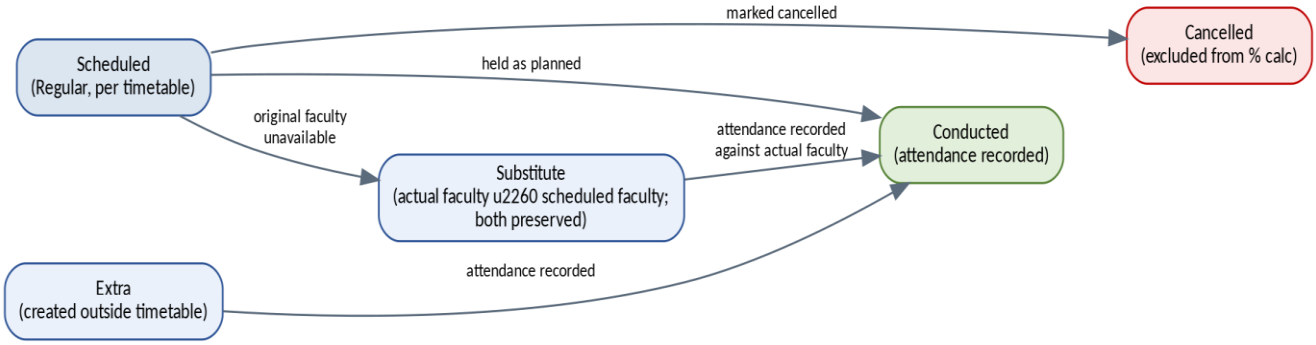


Figure 5 — Lecture type/state model (Regular, Substitute, Extra, Cancelled)

4.4.3 Functional Requirements

ID	Requirement
FR-LECT-001	The system shall represent every lecture/session with, at minimum: class, course, scheduled faculty, actual conducting faculty, date, start time, end time, lecture type, and classroom (where applicable).
FR-LECT-002	The system shall support three lecture types: Regular, Substitute, and Extra, and shall make the type visible wherever the lecture is displayed.
FR-LECT-003	For a Substitute lecture, the system shall record and preserve both the originally scheduled faculty member and the actual conducting faculty member as distinct data elements.
FR-LECT-004	The system shall associate attendance records with the actual conducted lecture rather than the originally scheduled faculty assignment.
FR-LECT-005	The system shall allow an authorized user to create an Extra lecture by specifying class, course, faculty, date, time, and classroom (where applicable).
FR-LECT-006	The system shall include Extra lectures in relevant scheduling views, attendance history, and attendance percentage calculations.
FR-LECT-007	The system shall allow an authorized user to mark a scheduled lecture as cancelled, and shall exclude cancelled lectures from attendance percentage calculations.
FR-LECT-008	The system shall retain the relationship between a Substitute lecture and its original scheduled lecture entry rather than creating a disconnected attendance record.
FR-LECT-009	The system shall display the lecture type (Regular, Substitute, Extra) in a student's and faculty member's attendance history.

4.5 Timetable Management

4.5.1 Description and Priority

Provides timetable data to Students, Faculty, HOD, and Admin, including both the complete/overall weekly timetable and a day-wise timetable for a user-selected day. Timetable access is not restricted to the current day. The feature also detects scheduling conflicts among faculty, classrooms, and classes.

Priority: High.

4.5.2 Stimulus/Response Sequences

- An authorized user requests the complete timetable and the system displays the full weekly schedule relevant to that user's role (student's class, faculty's assigned classes, or administrative scope).
- An authorized user selects a specific day (e.g., Monday) and the system displays only the lectures scheduled for that day.
- Admin creates or edits a timetable entry specifying class, course, faculty, classroom, day, and time; the system checks for conflicts before saving.
- The system detects that a proposed timetable entry overlaps an existing faculty assignment, classroom assignment, or class schedule, and warns the user before the entry is saved.
- An Extra or Substitute lecture is created; the system reflects it appropriately within the relevant timetable views.

The conflict-detection sequence below is shown as a decision flow, since it is one of the more intricate validation paths in the system — a single entry must clear three independent overlap checks before it can be saved.

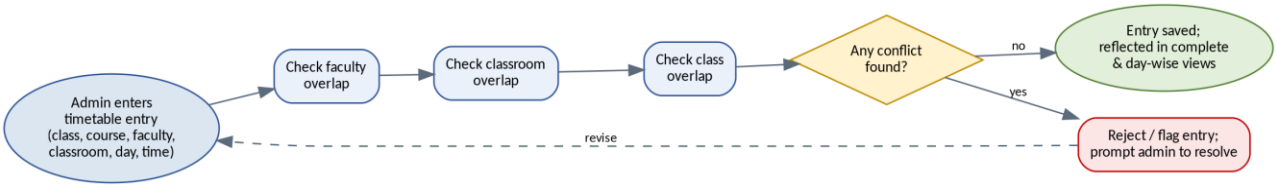


Figure 6 — Timetable entry creation and conflict-detection flow

4.5.3 Functional Requirements

ID	Requirement
FR-TT-001	The system shall allow an authorized user to view the complete/overall timetable relevant to their role.
FR-TT-002	The system shall allow an authorized user to select a specific day and view the day-wise timetable for that day.
FR-TT-003	The system shall not restrict timetable viewing to only the current day.
FR-TT-004	Each timetable entry shall include, at minimum: day/date, start time, end time, class, course, faculty, and classroom.

ID	Requirement
FR-TT-005	The system shall provide a student timetable view scoped to the student's assigned class.
FR-TT-006	The system shall provide a faculty timetable view scoped to the classes/courses assigned to that faculty member.
FR-TT-007	The system shall provide a class-level timetable view for use by Faculty and Admin.
FR-TT-008	The system shall provide administrative timetable views appropriate to Admin's system-wide scope.
FR-TT-009	The system shall detect and flag an attempted timetable entry that overlaps an existing assignment for the same faculty member.
FR-TT-010	The system shall detect and flag an attempted timetable entry that overlaps an existing assignment for the same classroom.
FR-TT-011	The system shall detect and flag an attempted timetable entry that overlaps an existing schedule for the same class.
FR-TT-012	The system shall reflect Extra and Substitute lectures appropriately within the relevant timetable views without displacing the underlying Regular schedule record.

4.6 Classroom Availability

4.6.1 Description and Priority

Allows authorized users to identify which classrooms are free during a requested date and time range, by comparing the request against existing timetable and lecture data, including Extra and Substitute lectures.

Priority: Medium.

4.6.2 Stimulus/Response Sequences

- An authorized user specifies a date and a time range and requests a list of free classrooms.
- The system compares the requested time range against all scheduled lectures (Regular, Substitute, and Extra) for each classroom.
- The system returns each classroom's status as Available or Occupied for the requested period.
- A classroom with an adjacent, non-overlapping lecture is correctly reported as Available for the requested slot.

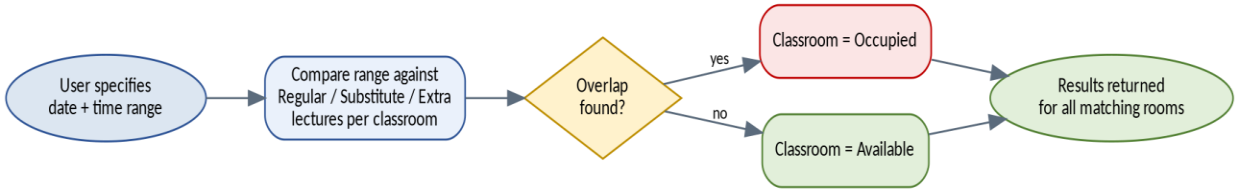


Figure 7 — Classroom availability search flow

4.6.3 Functional Requirements

ID	Requirement
FR-ROOM-001	The system shall allow an authorized user to request classroom availability for a specified date and time range.
FR-ROOM-002	The system shall determine a classroom's availability by comparing the requested time range against all Regular, Substitute, and Extra lectures scheduled in that classroom.
FR-ROOM-003	The system shall report a classroom as Occupied when a scheduled lecture in that classroom overlaps the requested time range, and Available otherwise.
FR-ROOM-004	The system shall not report adjacent, non-overlapping lectures as a scheduling conflict.
FR-ROOM-005	The system shall display availability results for all classrooms matching the search criteria in a single response.

4.7 Study Material Management

4.7.1 Description and Priority

Allows Faculty to upload and organize study material for the courses they teach, and allows Students to access material for their enrolled courses.

Priority: Medium.

4.7.2 Stimulus/Response Sequences

- Faculty uploads a study material file (e.g., PDF, PPT/PPTX, DOC/DOCX, image, or link) and associates it with a course and, optionally, a unit/topic.
- A Student opens a course and views the list of available study material organized by unit/topic.
- A Student downloads or opens a study material item.
- Faculty updates or removes a previously uploaded study material item.

4.7.3 Functional Requirements

ID	Requirement
FR-MAT-001	The system shall allow Faculty to upload study material associated with a course they are authorized to teach.
FR-MAT-002	The system shall allow study material to be organized by unit/topic within a course.
FR-MAT-003	The system shall support common academic file types for study material, including PDF, PPT/PPTX, DOC/DOCX, images, and educational links.
FR-MAT-004	The system shall allow Students to view and access study material for courses associated with their assigned class.
FR-MAT-005	The system shall allow Faculty to update or remove study material they previously uploaded for their authorized courses.

ID	Requirement
FR-MAT-006	The system shall prevent Students from accessing study material for courses not associated with their assigned class.

4.8 Assignment Management

4.8.1 Description and Priority

Enables Faculty to create, publish, and grade assignments, and Students to view, submit, and track feedback on assignments for their enrolled courses.

Priority: High.

4.8.2 Stimulus/Response Sequences

- Faculty creates an assignment specifying title, description, deadline, maximum marks, and optional attachments, and publishes it to a course.
- A Student views a published assignment and, before the deadline, uploads a submission.
- A Student submits after the deadline; the system records the submission as Late.
- Faculty reviews submissions, assigns marks, and provides feedback; the submission state changes to Graded.
- A Student views their own marks, feedback, and submission status for the assignment.

The states an assignment submission passes through are shown below as a state diagram, complementing the state list in FR-ASG-005.

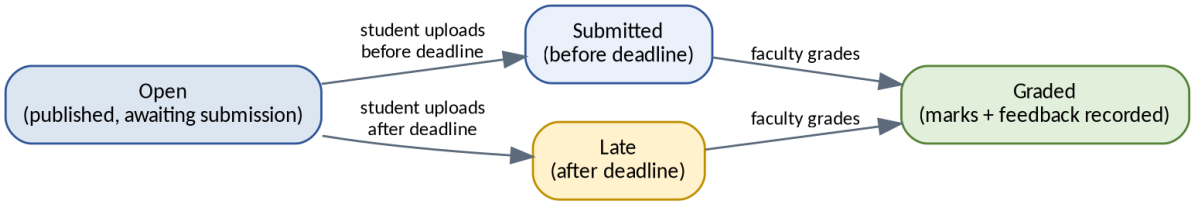


Figure 8 — Assignment submission state diagram (Open → Submitted/Late → Graded)

4.8.3 Functional Requirements

ID	Requirement
FR-ASG-001	The system shall allow Faculty to create an assignment specifying title, description, deadline, maximum marks, and optional attached files.
FR-ASG-002	The system shall allow Faculty to publish an assignment to the students of an assigned class/course.
FR-ASG-003	The system shall allow Students to view published assignments for their enrolled courses.

ID	Requirement
FR-ASG-004	The system shall allow Students to upload a submission for an assignment and shall record the submission timestamp.
FR-ASG-005	The system shall support assignment/submission states of at least: Open, Submitted, Late, and Graded.
FR-ASG-006	The system shall mark a submission as Late when it is uploaded after the specified deadline, while still accepting the submission unless the assignment is configured to disallow late submission.
FR-ASG-007	The system shall allow Faculty to view all submissions for an assignment they created.
FR-ASG-008	The system shall allow Faculty to assign marks and provide feedback text for a submission.
FR-ASG-009	The system shall allow a Student to view only their own submission status, marks, and feedback.
FR-ASG-010	The system shall prevent a Student from viewing another student's submission, marks, or feedback.

4.9 Quiz and Test Management

4.9.1 Description and Priority

Enables Faculty to create online quizzes/tests composed of objective questions (MCQ, True/False) and enables Students to attempt them within defined availability windows and duration limits. Objective questions are evaluated automatically and deterministically.

Priority: High.

4.9.2 Stimulus/Response Sequences

- Faculty creates a quiz specifying title, instructions, questions, marks per question, duration, availability window, and attempt rules.
- A Student starts an available quiz within the availability window; the system enforces the configured duration and attempt limit.
- A Student answers questions and submits the quiz, or the quiz auto-submits upon timeout/expiry.
- The system automatically evaluates objective answers, calculates the score, and records the attempt and answers.
- A Student views their quiz result once results are released by Faculty or by system configuration.

The end-to-end attempt flow, including the availability/attempt-limit gate and the submit-vs-timeout branch, is shown below.

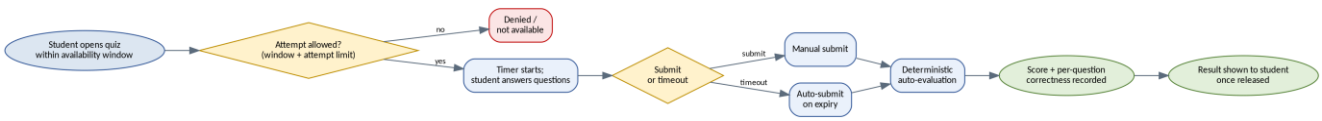


Figure 9 — Quiz attempt flow, from start through deterministic evaluation

4.9.3 Functional Requirements

ID	Requirement
FR-QUIZ-001	The system shall allow Faculty to create a quiz specifying title, instructions, questions, marks, duration, availability window, and attempt rules.
FR-QUIZ-002	The system shall support MCQ and True/False objective question types.
FR-QUIZ-003	The system shall allow Students to view quizzes available to their enrolled courses within the configured availability window.
FR-QUIZ-004	The system shall allow a Student to start, answer, and submit a quiz attempt.
FR-QUIZ-005	The system shall enforce the configured quiz duration and shall auto-submit the attempt upon timeout/expiry.
FR-QUIZ-006	The system shall automatically and deterministically evaluate objective question answers and calculate the resulting score.
FR-QUIZ-007	The system shall record each quiz attempt, including submitted answers, per-question correctness, and topic/question-level performance where configured.
FR-QUIZ-008	The system shall enforce the configured attempt-limit rules for a given quiz (e.g., single attempt or multiple attempts).
FR-QUIZ-009	The system shall allow Students to view their quiz results once results are released.
FR-QUIZ-010	The system shall prevent a Student from viewing quiz questions or another student's answers prior to or outside their own authorized attempt.

4.10 Results and Academic Performance

4.10.1 Description and Priority

Aggregates assignment and quiz outcomes into course-level and overall academic performance information viewable by Students for themselves, by Faculty for their classes/courses, and by HOD/Admin according to their broader authorization.

Priority: High.

4.10.2 Stimulus/Response Sequences

- A Student views their results screen and the system displays assignment marks, quiz marks, percentages, and course-level performance.
- Faculty views class/course-level performance for a course they teach.
- HOD views department-level academic performance information within their authorization.
- Admin views system-level academic performance information within their authorization.

4.10.3 Functional Requirements

ID	Requirement
FR-RESULT-001	The system shall allow a Student to view their own assignment marks, quiz marks, percentages, and course-level performance.
FR-RESULT-002	The system shall prevent a Student from viewing another student's academic results.
FR-RESULT-003	The system shall allow Faculty to view aggregated performance for classes/courses they are authorized to teach.
FR-RESULT-004	The system shall allow HOD to view department-level academic performance information consistent with their HOD permissions.
FR-RESULT-005	The system shall allow Admin to view system-level academic performance information consistent with their administrative authorization.
FR-RESULT-006	The system shall display performance trend information (e.g., across assignments/quizzes over time) where sufficient data exists for a course.

4.11 Dashboards and Academic Analytics

4.11.1 Description and Priority

Provides each user role with a landing dashboard summarizing the information most relevant to that role: attendance, academic performance, upcoming work, notifications, and role-specific administrative or analytical summaries.

Priority: High.

4.11.2 Stimulus/Response Sequences

- A Student logs in and views a dashboard summarizing attendance, low-attendance alerts, academic performance, upcoming assignments/quizzes, recent results, timetable, notifications, and calendar information.
- A Faculty member logs in and views a dashboard summarizing assigned classes/courses, attendance statistics, defaulters, assignment/quiz information, timetable, notifications, and announcements.
- An HOD-permissioned Faculty member views additional department-level analytics (attendance, performance, defaulters, pending grievances) alongside their standard Faculty dashboard.
- Admin logs in and views a dashboard summarizing users, classes, faculty, students, courses, classrooms, timetable, grievances, and system-level statistics.

4.11.3 Functional Requirements

ID	Requirement
FR-DASH-001	The system shall provide a Student dashboard summarizing attendance statistics, low-attendance alerts, academic performance, upcoming assignments and quizzes, recent results, timetable information, notifications, and relevant calendar entries.

ID	Requirement
FR-DASH-002	The system shall provide a Faculty dashboard summarizing assigned classes/courses, attendance statistics, defaulters, assignment and quiz information, timetable information, notifications, and announcements.
FR-DASH-003	The system shall provide additional HOD dashboard content, visible only to users holding the HOD permission, covering department-level attendance, academic performance, defaulter information, and grievances requiring attention.
FR-DASH-004	The system shall provide an Admin dashboard summarizing users, classes, faculty, students, courses, classrooms, timetable status, grievances, and relevant system-level academic/attendance statistics.
FR-DASH-005	The system shall restrict every dashboard element to only the information the viewing user is authorized to see.

4.12 Announcements and Notifications

4.12.1 Description and Priority

Provides two conceptually distinct communication mechanisms: Announcements, which are published messages targeted to an audience (class, course, college, or event audience), and Notifications, which are system-generated alerts tied to specific events or actions.

Priority: Medium.

4.12.2 Stimulus/Response Sequences

- An authorized user (Faculty, Admin, or Council Head via automatic event publication) publishes an announcement targeted to a class, course, the college, or an event audience.
- A targeted user views published announcements relevant to them.
- A triggering event occurs (e.g., new assignment, approaching deadline, new quiz, released result, low attendance, timetable change, new announcement, or new event); the system generates a notification for the relevant user(s).
- A user views their notifications and can distinguish them from announcements.

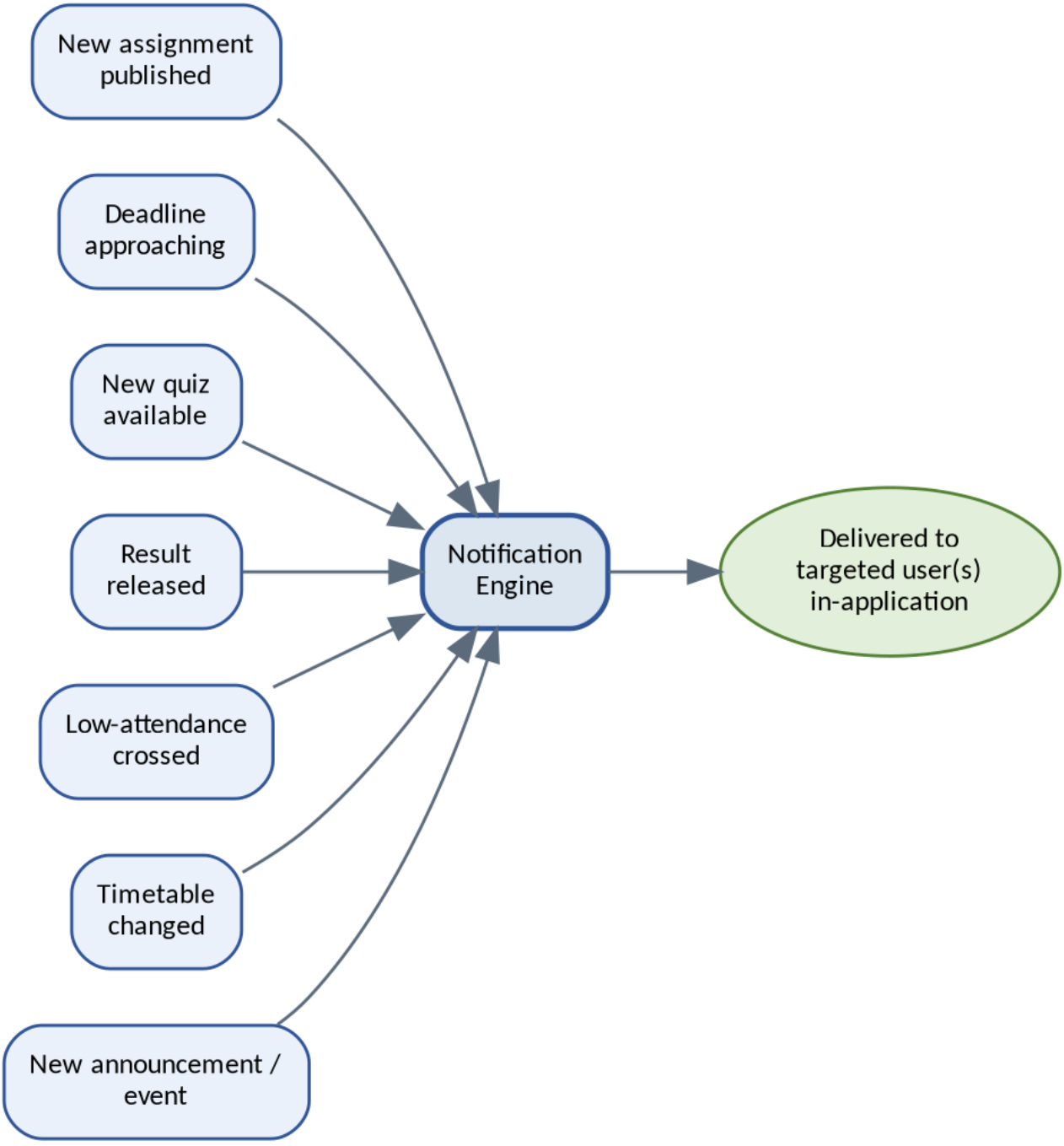


Figure 10 — Notification triggers and delivery flow

4.12.3 Functional Requirements

ID	Requirement
FR-NOTIF-001	The system shall allow authorized users to publish an announcement targeted to a class, course, the whole college, or an event audience.

ID	Requirement
FR-NOTIF-002	The system shall display announcements to users within the announcement's targeted audience.
FR-NOTIF-003	The system shall treat announcements and notifications as distinct concepts and shall present them separately to the user.
FR-NOTIF-004	The system shall automatically generate a notification for a user when a relevant triggering event occurs, including at minimum: new assignment, approaching assignment deadline, new quiz, released quiz result, released assignment result, low-attendance alert, timetable change, new announcement, and new event.
FR-NOTIF-005	The system shall allow a user to view their notification history.
FR-NOTIF-006	The system shall restrict announcement and notification visibility to users within the intended audience.

4.13 Calendar Management

4.13.1 Description and Priority

Provides a calendar view filterable by Events, Tests, and Holidays, displaying entries appropriate to the viewing user's permissions. Assignments and regular lectures are not mandatory calendar filter categories.

Priority: Medium.

4.13.2 Stimulus/Response Sequences

- A user opens the calendar and views entries across the default set of categories, filtered to what they are authorized to see.
- A user filters the calendar by Events only, or by Tests only, or by Holidays only, or by a combination of these.
- A Council Head publishes a college event and the system automatically adds a corresponding calendar entry under the Events filter.
- A Faculty-created quiz/test with a scheduled availability window appears under the Tests filter.

4.13.3 Functional Requirements

ID	Requirement
FR-CAL-001	The system shall provide a calendar view of entries relevant to the authenticated user.
FR-CAL-002	The system shall support filtering the calendar by the categories Events, Tests, and Holidays.
FR-CAL-003	The system shall not require Assignments or Regular lectures to be mandatory calendar filter categories.
FR-CAL-004	The system shall display only calendar entries the viewing user is authorized to see, based on their role and the entry's visibility.

ID	Requirement
FR-CAL-005	The system shall automatically create a calendar entry under the Events filter when a Council Head publishes an event.

4.14 College Council Event Management

4.14.1 Description and Priority

Allows Council Heads to create and manage events for their council, with Council-only or College-wide visibility. Publishing an event automatically creates a corresponding calendar entry and announcement, without requiring separate manual steps.

Priority: Medium.

4.14.2 Stimulus/Response Sequences

- A Council Head creates an event specifying title, description, date, time, venue, organizer/council, registration link, and visibility (Council-only or College-wide).
- The Council Head publishes the event; the system automatically creates a calendar entry and generates an announcement from the event information.
- A user with appropriate visibility views the event and sees its full details, including the registration link.
- A user clicks the registration link and is directed to the external registration system, if provided.
- A Council Head edits or removes an event they previously created; the associated calendar entry and announcement are updated or removed accordingly.

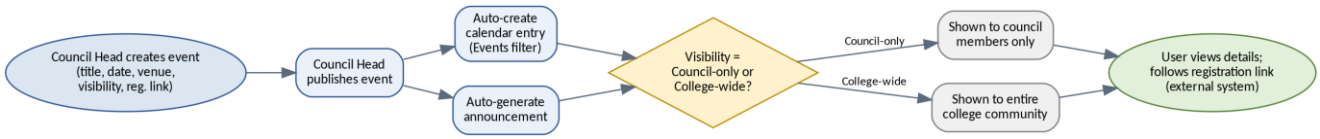


Figure 11 — Council event creation, publication, and visibility flow

4.14.3 Functional Requirements

ID	Requirement
FR-EVENT-001	The system shall allow a Council Head to create an event specifying title, description, date, time, venue, organizer/council, and an optional external registration link.
FR-EVENT-002	The system shall allow a Council Head to set an event's visibility to either Council-only or College-wide.
FR-EVENT-003	The system shall display a Council-only event only to members of the relevant council/audience.
FR-EVENT-004	The system shall display a College-wide event to the broader college community.

ID	Requirement
FR-EVENT-005	The system shall automatically create a corresponding calendar entry when a Council Head publishes an event, without requiring a separate manual calendar action.
FR-EVENT-006	The system shall automatically generate an announcement from the event information when a Council Head publishes an event, without requiring a separate manual announcement action.
FR-EVENT-007	The system shall display full event details — date, time, venue, organizer, description, and registration information — when a user selects an event.
FR-EVENT-008	The system shall allow an event's registration link to direct users to an external registration system; the system is not required to implement event registration itself.
FR-EVENT-009	The system shall allow a Council Head to edit or remove events they created, restricted to events belonging to their own council.

4.15 Discussion Forums

4.15.1 Description and Priority

Provides a per-course discussion forum where Students and Faculty of that course can post and reply to academic discussion topics, restricted to authorized course/class participants.

Priority: Low.

4.15.2 Stimulus/Response Sequences

- A Student in a course opens the course's discussion forum and posts a new discussion topic/question.
- Another Student or Faculty member replies to an existing discussion topic.
- Faculty moderates a discussion (e.g., removes an inappropriate post) within a course they teach.
- A user not enrolled in or assigned to the course attempts to access its forum and is denied.

4.15.3 Functional Requirements

ID	Requirement
FR-FORUM-001	The system shall provide a discussion forum associated with each course.
FR-FORUM-002	The system shall allow Students enrolled in a course's associated class to create discussion topics and reply within that course's forum.
FR-FORUM-003	The system shall allow Faculty assigned to a course to participate in and moderate that course's forum.
FR-FORUM-004	The system shall restrict access to a course's forum to students and faculty authorized for that course/class.

4.16 Feedback and Grievance Management

4.16.1 Description and Priority

Allows Students to submit and track feedback or grievances, and allows Admin and HOD (who may also be teaching Faculty) to review, respond to, and resolve them through a defined status lifecycle.

Priority: Medium.

4.16.2 Stimulus/Response Sequences

- A Student submits a grievance specifying category, title/subject, description, and an optional attachment.
- The system assigns the grievance an initial status of SUBMITTED and routes it to the appropriate Admin/HOD authority.
- Admin or HOD reviews an assigned grievance, updates its status through the lifecycle (ASSIGNED, UNDER REVIEW, IN PROGRESS, RESOLVED, CLOSED), and provides responses/remarks.
- A Student views the status and responses/updates for their own previously submitted grievances.
- A user without authorization attempts to view another user's grievance and is denied.

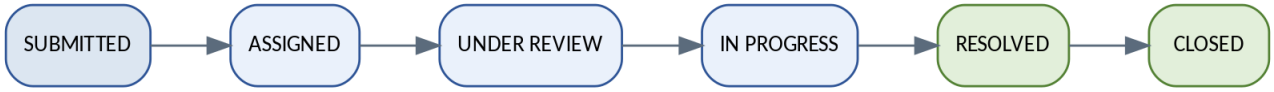


Figure 12 — Grievance status lifecycle

4.16.3 Functional Requirements

ID	Requirement
FR-GRV-001	The system shall allow a Student to submit a grievance specifying category, title/subject, description, and an optional attachment.
FR-GRV-002	The system shall support a grievance status lifecycle of at least: SUBMITTED, ASSIGNED, UNDER REVIEW, IN PROGRESS, RESOLVED, and CLOSED.
FR-GRV-003	The system shall allow Admin and HOD-permissioned users to view grievances assigned to them.
FR-GRV-004	The system shall allow Admin and HOD-permissioned users to update a grievance's status and provide response/remark text.
FR-GRV-005	The system shall allow a Student to view the current status and responses for grievances they submitted.
FR-GRV-006	The system shall restrict visibility of a grievance's content to the submitting Student and the Admin/HOD user(s) authorized to handle it.
FR-GRV-007	The system shall support a user who holds both Faculty and HOD permissions handling grievances through their HOD permission without requiring a separate account.

4.17 AI-Generated Test Performance Reports

4.17.1 Description and Priority

After a quiz/test has been deterministically evaluated, this feature sends the authoritative score and topic-wise performance data to an external AI/LLM service to generate an advisory report identifying strong areas, weak areas, and suggested revision topics. The AI output is strictly advisory and never determines or alters the official academic score.

Priority: Medium.

4.17.2 Stimulus/Response Sequences

- A quiz attempt is deterministically evaluated and an official score and topic-wise performance breakdown are produced.
- The system submits the score and topic-wise performance data to the configured AI/LLM service and requests an analysis.
- The AI service returns a performance summary identifying strong areas, weak areas, and suggested revision topics; the system stores and displays this as an advisory report alongside the official result.
- The AI service is unavailable or returns an error; the system still displays the official quiz result and indicates that the AI-generated report is temporarily unavailable.
- A Student views their AI-generated test performance report alongside their official quiz result.



Figure 13 — Quiz evaluation and AI advisory-report generation flow

4.17.3 Functional Requirements

ID	Requirement
FR-AI-001	The system shall generate an AI-based advisory performance report only after the official quiz/test score has been deterministically calculated.
FR-AI-002	The system shall provide the deterministic score and topic-wise performance data as input to the configured external AI/LLM service.
FR-AI-003	The AI-generated report shall include, at minimum, an overall performance summary, identified strong areas, identified weak areas, and suggested revision topics.
FR-AI-004	The system shall never allow the AI service to determine, override, or alter the official academic score of a quiz/test.
FR-AI-005	The system shall display the official quiz/test result independently of the availability of the AI-generated report.
FR-AI-006	The system shall indicate to the Student when the AI-generated report is temporarily unavailable, without blocking access to the official result.
FR-AI-007	The system shall allow a Student to view only their own AI-generated test performance report.

ID	Requirement
FR-AI-008	The system shall not present the AI-generated report as a general-purpose AI chatbot or tutoring feature; its scope is limited to test performance analysis.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The following targets apply under normal operating load (defined as concurrent usage consistent with a single mid-sized college's active student and faculty population) and assume a client with a typical broadband or campus network connection:

ID	Requirement
NFR-PERF-001	The system shall complete user login authentication and return the role-appropriate dashboard within 3 seconds under normal operating load.
NFR-PERF-002	The system shall render dashboard summary data (Section 4.11) within 3 seconds of dashboard load under normal operating load.
NFR-PERF-003	The system shall persist a saved attendance record for a full class roster within 3 seconds under normal operating load.
NFR-PERF-004	The system shall accept and acknowledge a quiz submission within 2 seconds under normal operating load, excluding client-side upload time for any attached files.
NFR-PERF-005	The system shall return complete or day-wise timetable data within 2 seconds under normal operating load.
NFR-PERF-006	The system shall return classroom-availability search results within 3 seconds under normal operating load.
NFR-PERF-007	The system shall return calendar data for the requested filter(s) within 2 seconds under normal operating load.
NFR-PERF-008	For standard CRUD-style API operations not otherwise specified above, the system shall return a response within 2 seconds at the 95th percentile under normal operating load.
NFR-PERF-009	The AI-generated test performance report (Section 4.17) shall be produced within 30 seconds of a quiz's official evaluation under normal operating conditions for the external AI service; if this target is exceeded, the report shall be marked as pending rather than blocking display of the official result.

5.2 Safety Requirements

SmartClass is an academic information system and does not control physical equipment or processes that pose direct physical-safety risk. “Safety” in this context is interpreted as protection against loss, corruption, or unintended disclosure of academic and personal records that could cause institutional or reputational harm.

ID	Requirement
NFR-SAFE-001	The system shall prevent the permanent loss of saved attendance, assignment submission, or quiz attempt data due to a single point of failure, through the backup/reliability measures specified in Section 5.4 (Reliability).

ID	Requirement
NFR-SAFE-002	The system shall prevent a Student from performing any action that would alter their own or another student's official attendance or academic result data (see BR-015, BR-016).
NFR-SAFE-003	The system shall log all changes to official attendance and result records with sufficient detail (who, what, when) to support institutional audit and dispute resolution.

5.3 Security Requirements

ID	Requirement
NFR-SEC-001	The system shall store all user passwords using a one-way cryptographic hash function with per-user salt; plaintext password storage is prohibited (see FR-AUTH-004, BR-028).
NFR-SEC-002	The system shall require successful authentication for every function other than login and public informational pages.
NFR-SEC-003	The system shall enforce role-based access control on every API operation and data query, denying and logging unauthorized access attempts (see FR-AUTH-008, FR-AUTH-010).
NFR-SEC-004	The system shall use time-limited session tokens and shall invalidate a token upon logout, expiry, or password change.
NFR-SEC-005	The system shall validate and sanitize all user-supplied input on the server side to prevent injection-style vulnerabilities.
NFR-SEC-006	The system shall restrict access to uploaded files (study material, assignment attachments, submissions, grievance attachments) to users authorized for the associated course, class, or grievance.
NFR-SEC-007	The system shall transmit all data between client and server, and between the server and the external AI service, over an encrypted channel (HTTPS/TLS).
NFR-SEC-008	The system shall maintain an audit log of sensitive administrative actions, including account creation, role/permission changes, and grievance status changes.
NFR-SEC-009	The system shall lock an account after a configurable number of consecutive failed login attempts, per FR-AUTH-006.

5.4 Software Quality Attributes

Privacy

The system shall protect student personal information, attendance, marks, assignment submissions, quiz results, grievances, and faculty information from access by any user not explicitly authorized to view that data, consistent with the role-scoping requirements in Section 4 and the business rules in Section 5.5.

Reliability

The system shall reliably preserve attendance records, assignment submissions, quiz attempts/results, timetable information, event information, and grievances through routine database backup and

transaction-integrity practices, such that a single component failure does not result in silent data loss for a successfully acknowledged user action.

Availability

The system shall target a monthly uptime of at least 99.0% during the academic term, excluding scheduled maintenance windows communicated to users in advance, reflecting the availability expectations reasonable for a college academic information system that is not life-safety critical.

Usability

The system shall present a responsive interface that is consistent, understandable, and easy to navigate for Student, Faculty, Admin, HOD, and Council Head users of varying technical proficiency, and shall be fully usable through mobile web browsers, consistent with the responsive-web-application constraint in Section 2.5.

Maintainability

The system shall be structured to support modular development and a clear separation of responsibilities across its major functional areas, enabling independent extension of individual features (e.g., adding a new quiz question type) without requiring changes throughout the codebase. This SRS does not prescribe a specific architecture pattern; that decision belongs to the downstream Software Architecture and Design document.

Testability

Every functional requirement in Section 4 shall be stated so that it is verifiable through unit testing, integration testing, browser automation, and/or behavior-driven (BDD) scenario testing. The requirement set shall additionally support mutation testing of the deterministic evaluation logic described in Section 4.9 and Section 4.17, given the correctness sensitivity of official academic scoring.

5.5 Business Rules

The following business rules describe institutional operating principles that constrain or motivate the functional requirements in Section 4. They are not, in themselves, functional requirements, but each implies one or more of the requirements listed above.

ID	Requirement
BR-001	Only authorized administrators can create/manage institutional user accounts.
BR-002	Students are assigned to classes by Admin.
BR-003	Faculty are assigned to appropriate classes/courses by authorized administration.
BR-004	Students can access only their assigned class and authorized courses.
BR-005	Faculty can access only assigned classes/courses.
BR-006	Attendance is manually recorded by authorized faculty/users.
BR-007	Attendance is stored per lecture/session.
BR-008	Regular, Substitute and Extra lectures must be distinguishable.

ID	Requirement
BR-009	Substitute lectures must preserve the original scheduled faculty and actual conducting faculty where applicable.
BR-010	Extra lectures must be reflected correctly in attendance history.
BR-011	Attendance percentage is calculated using valid conducted lectures.
BR-012	Low-attendance alerts are generated when configured thresholds are crossed.
BR-013	Students can view their own attendance history.
BR-014	Faculty can view attendance for authorized classes/courses.
BR-015	Students cannot modify official attendance records.
BR-016	Students cannot modify official academic results.
BR-017	Objective quiz results are calculated by deterministic system logic.
BR-018	AI does not determine official academic scores.
BR-019	AI-generated test reports are advisory.
BR-020	A Council Head can manage events belonging to their council.
BR-021	Council events can be Council-only or College-wide.
BR-022	Publishing a council event automatically creates the relevant calendar entry.
BR-023	Publishing a council event automatically creates the relevant announcement.
BR-024	Students can submit grievances.
BR-025	Admin/HOD can handle grievances according to their permissions.
BR-026	HOD may also have Faculty responsibilities, without requiring a separate account.
BR-027	Users cannot access unauthorized personal, academic, attendance, or grievance information.
BR-028	Passwords must not be stored in plaintext.
BR-029	Classroom availability is determined from timetable/lecture conflicts.
BR-030	Both complete timetable and day-wise timetable views must be available to users with timetable access.

6. Other Requirements

No database schema, legal/regulatory, internationalization, or software-reuse requirements beyond those already stated in Sections 2, 3, and 5 have been identified for Version 1.0. Any institution-specific regulatory requirement (e.g., a national student-data-protection regulation) that applies to a specific deployment should be captured as an addendum to this SRS once the deploying institution is known; in the absence of that information, this is recorded as TBD-001 in Appendix C.

Appendix A: Glossary

Term	Definition
SmartClass	The integrated academic and classroom management platform specified by this document.
Admin	The primary institutional/system administrator, holding the highest level of system authority.
Faculty	A teaching staff member who delivers courses to one or more classes.
Student	A learner enrolled in a class and its associated courses.
HOD	Head of Department; an additional permission set that may be granted to a Faculty user, rather than a mandatory separate account.
Council Head	A user responsible for creating and managing events on behalf of a specific college council.
Class	A student batch/division, e.g., “TE Computer Engineering – Division A.”
Course	An academic subject, e.g., “Software Architecture,” that may be associated with one or more classes.
Lecture / Session	A single scheduled teaching event associated with a class, course, faculty member, date, and time, against which attendance is recorded.
Regular Lecture	A lecture conducted as originally scheduled in the timetable.
Substitute Lecture	A scheduled lecture conducted by a faculty member other than the one originally assigned, with both the scheduled and actual faculty preserved.
Extra Lecture	An additional lecture conducted outside the normal timetable, recorded and included in scheduling and attendance calculations.
Attendance	A record of a student's Present/Absent status for a specific lecture/session.
Attendance Defaulter	A student whose attendance percentage has fallen below the configured institutional threshold.
Assignment	A graded academic task created by Faculty, with a deadline and maximum marks, for Student submission.
Quiz	An online, time-bound, objectively evaluated test created by Faculty for Students to attempt.
Quiz Attempt	A single instance of a Student attempting a quiz, including submitted answers and computed score.
Result	Academic outcome information derived from assignment marks and/or quiz scores.
Study Material	Course-related files or links uploaded by Faculty for Student access.
Announcement	A published message targeted to an audience such as a class, course, the college, or an event audience.

Term	Definition
Notification	A system-generated alert tied to a specific event or action, distinct from an announcement.
Calendar Event	An entry displayed on the calendar under the Events, Tests, or Holidays filter.
Council-only Event	An event whose visibility is restricted to the relevant council/audience.
College-wide Event	An event visible to the broader college community.
Grievance	A formal concern or complaint submitted by a Student and tracked through a defined status lifecycle.
AI-Generated Test Report	An advisory report produced by an external AI/LLM service, analyzing already-calculated quiz/test performance to identify strong and weak areas.
RBAC	Role-Based Access Control; the mechanism by which system access is restricted according to a user's assigned role(s).
SRS	Software Requirements Specification; this document.

Appendix B: Analysis Models

The following analysis models are identified as artifacts that will accompany this SRS during design. They are not embedded in this document; each is to be produced and maintained as a separate diagram/model file referenced from this SRS.

- System context diagram, showing SmartClass and its external actors/interfaces (Section 2.1, Section 3.3).
- Use case model covering the primary flows for Student, Faculty, Admin, and Council Head (Section 7 of the requirements-gathering source material).
- Level 0 (context-level) data flow diagram.
- Level 1 data flow diagram decomposing the major feature areas of Section 4.
- Level 2 data flow diagrams for the Attendance/Lecture management process (Section 4.3–4.4) and the Quiz/AI Report process (Section 4.9, Section 4.17), given their relative complexity.
- Domain/class model capturing the core entities: User (with Student/Faculty/Admin/Council Head/HOD roles), Class, Course, Lecture/Session, Attendance Record, Assignment, Submission, Quiz, Quiz Attempt, Result, Announcement, Notification, Calendar Entry, Event, Forum Post, and Grievance.
- Entity-relationship (ER) model for the PostgreSQL schema, to be produced in the downstream design phase.
- Sequence diagrams for the Substitute Lecture attendance flow (Section 9 of the requirements-gathering source material) and the AI-Generated Test Report flow (Section 4.17).
- Attendance workflow / state model showing Regular, Substitute, Extra, and Cancelled lecture states and their effect on attendance calculation.

Appendix C: To Be Determined List

ID	Requirement
TBD-001	Institution-specific regulatory or student-data-protection requirements applicable to the deploying college have not yet been identified and must be captured once the deploying institution is confirmed (see Section 6).
TBD-002	The delivery format for user documentation (in-application help, downloadable PDF guides, or both) has not yet been finalized (see Section 2.6).
TBD-003	The specific external AI/LLM vendor and model to be used for AI-generated test performance reports has not yet been selected; the requirement set in Section 4.17 is written to be vendor-independent (see Section 3.3).
TBD-004	The configurable institutional low-attendance threshold value(s) (e.g., 75% vs. 80%) have not yet been finalized and are expected to be set per deploying institution (see Section 2.7, FR-ATT-010).
TBD-005	The configurable password-lockout parameters (number of failed attempts, cool-down duration) have not yet been finalized (see FR-AUTH-006).